

Bruno Saboia de Albuquerque

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Professional Summary

Senior software engineer with over 20 years of experience delivering production systems for banking, insurance and international organisations — most recently conversational AI at AXA Group Operations and data analytics at the Bank for International Settlements. Works across the full lifecycle: C#, Python, SQL and TypeScript; DevOps, CI/CD and Kubernetes; and applied AI/ML, LLM and RAG systems. Economics graduate, published author and peer reviewer, experienced in distributed, multilingual teams across Europe and the Americas.

Work Experience

AXA Group Operations, Winterthur, CH

Conversational AI Engineer

October 2025 - July 2026

<https://axa.com>

Built the Python API and React front end of the LLM assistant used by AXA's claims handlers in Switzerland, and owned the internal developer tooling the rest of the team built on. Hardened the platform with PII redaction, GDPR compliance and cybersecurity controls to prepare its rollout beyond Switzerland into France and Spain.

Technologies: TypeScript, React, Python, LLM, RAG, Azure, Docker, GitHub

Bank for International Settlements, Basel, CH

Data Analytics Engineer

May 2024 - May 2025

<https://bis.org>

Built BIS GPT, the bank-wide internal LLM assistant, as one of its core engineers: Python API, React front end, developer tooling, and the OpenShift/ArgoCD delivery pipeline. Designed and operated the project's Azure stack on private OpenAI endpoints, enforcing the bank's document classification rules end to end.

Technologies: TypeScript, React, Python, LLM, RAG, Azure, OpenShift, Docker, ArgoCD, GitHub

Bank for International Settlements, Basel, CH

Business Analyst/Developer

May 2022 - May 2024

<https://bis.org>

Built and modernised the systems Risk Management used to measure the bank's credit, market, liquidity and counterparty risk, gathering requirements directly from risk analysts and implementing them end to end. Drove the department's engineering practice forward, containerising delivery onto OpenShift, running CI/CD on GitHub and Azure DevOps, and introducing a project-template CLI that became the default starting point for new work.

Technologies: C#, Git, Azure DevOps, Python, PowerShell, Bash, OpenShift, SQL, Octopus, ArgoCD

Clevertech, Remote (Brazil – US)

Senior Software Engineer

March 2022 - May 2022

<https://clevertech.biz>

Clevertech is an American IT company, providing consulting services for large customers around the globe.

Built the Postgres and Airflow pipelines behind a client's reporting dashboards, delivering the contract to completion.

Technologies: Python, Apache Airflow, Postgres

Tesseract, Remote (Brazil – Finland)

Full-stack developer

September 2021 - March 2022

<https://tesseractinvestment.com/>

Tesseract is a Finnish startup which borrows cryptocurrencies, matchmaking investors and lenders. It is Fin-FSA regulated digital asset innovator, backed by reputable investors, focusing on institutional brokerage and asset management.

Established the company's Infrastructure-as-Code platform on Pulumi and GitHub Actions, replacing a ClickOps, spawn-resources-as-needed practice with reproducible, version-controlled infrastructure.

Technologies: Pulumi, TypeScript, GitHub Actions

BEC Financial Technologies, Roskilde, Denmark

Senior Software Engineer

April 2021 - September 2021

<https://bec.dk>

BEC is a Danish fintech that provides services for medium-to-small banks. Founded in 1964, is one of the main back-bones of IT services for financial companies in the Nordic country.

Ran Kafka as an internal platform for the teams building BEC's payment systems, the event-streaming backbone serving small and medium-sized Danish banks. Brought DevOps discipline to it: GitOps for topics and other Kafka objects, and reproducible, traceable fixes in place of ad-hoc production commands.

Technologies: Kafka, Java, Bash, OpenShift, Git, CDC, Docker, Jenkins

Danske Bank, Copenhagen, Denmark

Senior Software Engineer

February 2019 - March 2021

<https://danskebank.dk>

Built SmartSearch, the corporate payment search product of Danske Bank's Payment Innovations department: change data capture off the DB2 payments table into Kafka Streams, indexed into Elasticsearch, keeping the proven COBOL/DB2 stack as the source of truth. Wrote the C# API behind its Angular UI, reconciled payments across both stores, moved the stack onto Kubernetes, and took over the project's transfer from Copenhagen to Aarhus during a hiring freeze.

Technologies: Kubernetes, DB2, C#, Jenkins, .NET Core, COBOL, Kafka, Java, Kotlin, Rust, ELK stack

Coinify ApS, Copenhagen, Denmark

Senior Software Engineer

August 2017 - January 2019

<https://coinify.com>

Coinify is a Nordic startup which enables merchants to accept cryptocurrencies in exchange for their goods and services, either in mortar-and-brick or online stores. The company also provides a cryptocurrency trading platform.

Built features across Coinify's customer-facing platform for buying and selling cryptocurrency by credit card and the internal tooling behind it, covering ad-hoc transfers and reconciliation. Co-owned and delivered the 3-D Secure implementation for credit-card payments with one other engineer.

Technologies: Node.js, TypeScript, Docker, AWS, PHP, HTML/CSS

AH Integration Solutions AS, Remote (Brazil – Norway)

Consultant

February 2017 - August 2017

<https://web.archive.org/web/20160208120600/http://ahint.no/>

AH Integrations was a Norwegian company working in the health sector.

Consulted on the design and build of a cost reporting platform, mentoring the intern working alongside me and preparing each day's work in advance to bridge our limited overlap in working hours.

Technologies: C#, WPF, Crystal Reports

ITOUCH TECH, Sarasota, United States

CTO

June 2015 - February 2017

<https://itouchtech.us>

ITOUCH TECH is an US-based company which provides FFLs with a digital way to fill ATF4473 forms.

Owned the technology of a four-person company as CTO, from architecture through hands-on delivery. Modernised the touchscreen kiosk that firearms dealers used to complete the federal ATF 4473 form without errors, which reported its own usage back for per-form billing, and replaced the manual update process with automated Octopus deployments as the store fleet expanded.

Technologies: C#, Windows Forms, WPF, Octopus

HBS, Fortaleza, Brazil

IT Specialist – contract

April 2014 - August 2014

<https://hbs.tv>

Founded in 1999, Host Broadcast Services (HBS) was originally established with the core mandate of being the Host Broadcaster (HB) for the 2002 FIFA World Cup Korea/Japan™; the organisation has since developed as a renowned host broadcast expert.

Ran the IT infrastructure and signal monitoring for HBS's Fortaleza venue at the 2014 FIFA World Cup, through the pre-tournament build-up and every match hosted there.

Technologies: AD, Windows Server, C#, VB, satellite comm, optical fiber

Predictive Science, Remote (Brazil – US)

Senior Software Engineer

January 2013 - March 2014

<https://predsci.com>

Predictive Science Inc. is an employee-owned company that delivers state-of-the-art scientific solutions.

Built the escalation dashboard used by Verizon's Executive Response Team, the group handling escalated, high-profile customer cases, giving them the data to judge which cases genuinely warranted escalation.

Technologies: Python, Kafka, C#

Upstream Systems, Athens, Greece

IT Specialist/Copywriter

July 2012 - December 2012

<https://www.upstreamsystems.com>

Upstream Systems is a Greek media company that helps mobile carriers to promote products and services to their customers.

Wrote Brazilian Portuguese copy for Brazilian mobile carriers, and built a VBA tool on the team's existing Excel workbook to track copywriting work.

Technologies: Excel, VBA

PGE-CE, Fortaleza, Brazil

Senior Software Engineer

March 2011 - December 2011

<https://pge.ce.gov.br>

Procuradoria Geral do Estado do Ceará (PGE-CE) is the state attorney central for the state of Ceará, in Brazil.

Served as one of the senior developers on the task force that moved the state attorney general's office to digital judicial filings, mandated by Brazil's higher courts, contributing to both the architecture and the implementation while guiding the junior developers on the team. Designed and solely owned PGE-C2, a cloud-based Android app that extended the web-based case management workflow into the attorneys' daily operations.

Technologies: Ruby, Rails, C#, Mono, Postgres, Solr, Android, Java

SecreNET, Fortaleza, Brazil

Senior Software Engineer

January 2009 - February 2011

<https://secrelnet.com.br>

SecreNET is an ISP of Fortaleza, Brazil. It was one of the pioneers of both dial-up and broadband technologies in the area.

Created Spider, a web tool that let the ISP's round-the-clock first-line support carry out hosting operations, such as creating accounts and sites or restarting a downed IIS site, that had until then required a second-line engineer holding machine-wide credentials. Adopted across most of the company's routine operations, it removed a recurring escalation path and the out-of-hours waits that came with it.

Technologies: .NET, C#, Java, C, Windows Development, WMI, VB, Linux, Hyper-V, Active Directory

Education

Universidade Federal do Ceará, Fortaleza, Brazil

Bachelor of Science, Economics

January 2013 - March 2016

<https://ufc.br>

Transcript of records: <https://saboia.it/assets/pdf/transcripts/2016-04-29-transcript-ufc.pdf>

Universität Liechtenstein, Vaduz, Liechtenstein

Exchange Student, Business Administration and IT

August 2015 - February 2016

<https://uni.li>

Transcript of records: <https://saboia.it/assets/pdf/transcripts/2016-02-17-transcript-liech.pdf>

Publications

- Over the Cliff: How Brazil's 2004–2016 Business Cycle Became Its Worst Recession in a Century (2019)
DOI: <https://doi.org/10.35297/qjae.010027>
- Understanding Bitcoins: Facts and Questions (2015)
DOI: <https://doi.org/10.5935/0034-7140.20150001>

Peer Reviews

Bachelor theses:

- Os efeitos do grau de Liberdade Econômica sobre o Índice de Desenvolvimento Humano (IDH) e o IDH ajustado à desigualdade, Renato Leitão (2022)
<https://saboia.it/assets/pdf/reviews/2022-02-15-renato-leitao.pdf>
- Bitcoins - uma moeda global e descentralizada, Raíssa Guerra (2022)
<https://saboia.it/assets/pdf/reviews/2022-02-15-raissa-guerra.pdf>
- Abertura comercial e produtividade: evidências empíricas de correlação nas últimas duas décadas, Saulo Queiroz (2021)
<https://saboia.it/assets/pdf/reviews/2021-08-26-saulo-queiroz.pdf>
- Mercado de criptoativos: conjuntura e desafios, Gabriel Radaelly (2021)
<https://saboia.it/assets/pdf/reviews/2021-08-25-gabriel-radaelly.pdf>
- A moeda digital do século XXI: um estudo de caso sobre o Bitcoin, Hemerson Braga (2021)
<https://saboia.it/assets/pdf/reviews/2022-07-07-hemerson-braga.pdf>

Papers:

- The Political Thought of José Osvaldo de Meira Penna, Gabriel F. Benzecry (2023)
<https://jls.mises.org/article/70305-the-political-thought-of-jose-osvaldo-de-meira-penna>

Awards and Honours

- SAP DemoJam Barcelona, 1st (November 2015)
<https://web.archive.org/web/20221225142713/https://www.uni.li/en/topics/information-systems/news/university-students-successful-in-world2019s-leading-sap-conference/>

Certifications

- Machine Learning (2013)
<https://www.coursera.org/api/legacyCertificates.v1/spark/statementOfAccomplishment/152~314678/pdf>
- Computational Investing, Part I (2013)

<https://www.coursera.org/api/legacyCertificates.v1/spark/statementOfAccomplishment/364~314678/pdf>

Skills

Programming Languages: C#, Python, SQL, TypeScript, JavaScript, Java, Kotlin, Rust, Ruby, PHP, C, PowerShell, Bash/Shell, VBA, VB, COBOL

AI & Machine Learning: LLM, RAG, Machine Learning, Azure OpenAI, Prompt Engineering, CUDA

Cloud & Containers: Azure, AWS, Kubernetes, OpenShift, Docker, Pulumi, Infrastructure as Code

DevOps & CI/CD: CI/CD, GitOps, Git, GitHub Actions, Azure DevOps, Jenkins, ArgoCD, Octopus Deploy

Data & Event Streaming: Kafka, Kafka Streams, Change Data Capture (CDC), Elasticsearch / ELK, Apache Airflow, PostgreSQL, DB2, Solr

Frameworks & Platforms: .NET / .NET Core, React, Angular, Node.js, Ruby on Rails, WPF, Windows Forms, Android, Mono, HTML/CSS

Systems & Networking: Linux, Windows Server, macOS, Active Directory, Hyper-V, WMI, Networking, Satellite communications, Optical fibre

Other: LaTeX, R / MATLAB / Octave, Crystal Reports

Languages

Portuguese (Native), English (Full professional proficiency), Spanish (Limited professional proficiency), German (Basic), Italian (Basic), Danish (Basic), Hebrew (Basic)

Interests

Sports: Go-karts, Martial arts

Music: Singing, Guitar, Piano

Research: Genealogy, History, Mathematics, Physics

Other: Motorbike, Photography, Cooking, Snooker, Hiking